

EPAM Data Engineering Interview

Today's Complete Question Set

Questions captured from today's interview discussion

1. Project / Introduction

1. Tell me about yourself.
2. Can you give me a brief overview of your overall experience?
3. Can you explain your recent project?
4. Can you explain your project, roles and responsibilities?
5. Was your previous project a migration project?
6. Where was the legacy code in your previous project?
7. You mentioned migrating from Informatica to Azure/cloud. Why did you prefer the target approach instead of Informatica Cloud?
8. What are the source systems you are using in your project?
9. How much experience do you have with Data Factory (ADF)?
10. What Azure services have you worked with?

2. Azure Data Factory / Architecture

1. Why did you use Azure Data Factory for ingestion? Why not do the complete flow in Databricks?
2. If you were the architect, would you choose ADF + Databricks or only Databricks? Why?
3. SQL Server has a 20-million-record table and you need a one-time load to another SQL Server table. Would you use ADF or Databricks? Why?
4. Apart from large data volume, what other reasons would make you choose ADF?
5. Both ADF and Databricks have notifications, triggers and monitoring. Why would you still choose ADF?
6. What activities are available in Azure Data Factory?
7. What is the purpose of the Get Metadata activity?
8. Can Get Metadata be used to check whether a specific file exists or not?
9. Have you used metadata/conditions to validate whether a file exists in a path?
10. A daily sales file is expected in ADLS, but it does not arrive. The ADF daily-triggered pipeline currently fails. How would you redesign it so it does not fail and instead sends a notification that the file has not arrived? Which activities would you use?
11. Multiple files arrive every day under sales/<today's date>/. How would you design the ADF pipeline to copy all files to the destination?
12. Can you design the multiple-file copy without using a ForEach activity?
13. How would you configure a pipeline when the date folder changes every day and you simply need to copy the files?
14. Customer files arrive in the same folder in CSV, Excel, JSON and ORC formats. How would you design the pipeline to process all these formats?

15. How would you implement CI/CD for Azure Data Factory?

3. Databricks / Medallion / Unity Catalog

1. Can you explain the Medallion architecture?
2. Sales files contain corrupted records, duplicate records and an incorrect schema, plus an aggregation requirement. What checks and transformations would you perform in each Medallion layer?
3. Power BI needs to consume your data. Would you use the Gold table directly or create views/other tables for reporting?
4. What are the advantages of Unity Catalog?
5. A metastore has five tables: three managed and two external. You need to migrate all five to Unity Catalog. How would you do it?
6. When migrating tables to Unity Catalog, what permissions are needed to create the tables and load the data?
7. Do you have experience with Delta Sharing?

4. Incremental Load / Keys

1. An on-prem SQL Server sales table is refreshed every day. A full load to the final target has performance issues. How would you implement an incremental approach?
2. What if the source table has no primary key, no secondary key, no load date, no timestamp and no obvious change-tracking column?
3. As a developer, before going back to the source team, what analysis would you perform?
4. If a required key is not available, what type of fix would you prefer?
5. Can you create a primary/surrogate key if the source does not provide one?
6. How would you create the key?
7. What is a hash key and where can you use it?
8. What are the different types of keys in SQL, such as primary key, unique key and business key?
9. What is the difference between a primary key, unique key and business key?

5. SQL / Joins / Data Quality

1. What are the different joins available in SQL?
2. Have you heard of LEFT SEMI JOIN and LEFT ANTI JOIN?
3. What is the difference between LEFT SEMI JOIN and LEFT ANTI JOIN?
4. Customer and sales files are joined using IDs. If the source reports corrupted data, how would you filter out corrupted records before they reach the required processing layer?
5. A customer file has five columns and a particular row has corrupted data in the fifth column. How would you exclude that row and capture it separately?
6. How can Spark help identify and capture corrupted records?
7. How would you handle corrupted/bad records during file ingestion without manually checking every record?

6. Spark / PySpark Performance

1. Two Delta tables have around 10 million and 5 million records. The join ran fine yesterday but is taking much longer today. How would you analyze the issue?
2. What fixes or optimizations would you suggest after finding the cause?
3. If the data/files are not organized properly or there are small files, what would you do?
4. If there is too much shuffle during the join, what optimization would you use?
5. If the tables have around 5 million and 10 million records, would you still consider a broadcast join?
6. Suppose executor memory is 5 GB and the two tables together are about 3 GB. Can you use a broadcast join?
7. Forget about simply saying which table is small or big. How do you decide which side should be broadcast?
8. Where would you use broadcast join in a real data-engineering use case?
9. What is the purpose/use case of caching or persistence?

7. Legacy SQL Migration / Query Redesign

1. The SQL team gives you a legacy SQL script of around 500 lines using around 15 tables, six inner joins, several left joins, filters and subqueries. Running it in Databricks has performance issues. How would you redesign it?
2. While redesigning the query, you notice duplicate records. How would you identify what is causing the duplication?
3. What type of fix would you apply instead of simply removing duplicates?
4. How would you use the execution plan and Spark UI to find the bottleneck?

8. Python

1. What is the purpose of a lambda function?
2. What are decorators?
3. What are generators?
4. Class A has several methods and Class B needs to use those methods. How can Class B call/use the methods of Class A?
5. Write a Python program to group anagrams from a list of words.
6. For the anagram problem, can you solve it without using sorted() or sort()?
7. Can you solve the anagram problem using loops instead of built-in sorting methods?

9. Agile / Sprint

1. How are stories/tasks planned in your team?
2. How many days is your sprint?
3. How long is sprint planning?
4. Do you have sprint ceremonies such as sprint review and retrospective?
5. What happens during the sprint review?

6. What happens during the sprint retrospective?

10. Quick Revision Priority

- ADF vs Databricks — when and why to use each
- Missing-file handling and multi-file ADF pipelines
- Medallion architecture with real data-quality checks
- Unity Catalog: managed/external tables and permissions
- Incremental loading when the source has no key or timestamp
- SQL joins, especially LEFT SEMI and LEFT ANTI
- Duplicate rows caused by one-to-many/many-to-many joins
- Spark join performance, shuffle, broadcast join and Spark UI
- Legacy SQL redesign and performance tuning
- Corrupt-record handling during ingestion
- Python OOP, lambda, decorators, generators and anagram coding
- Agile sprint planning, review and retrospective